

IBM Qiskit Fall Fest 2026

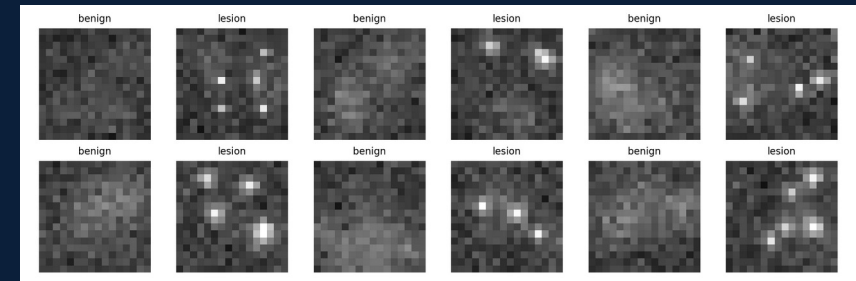
Updated: CV + noise + claim guardrails

Q-BioDiag

Quantum AI for Early Biomedical Diagnostics

4-qubit quantum-kernel classifier for benign vs lesion image patches, plus diagnostic case retrieval.

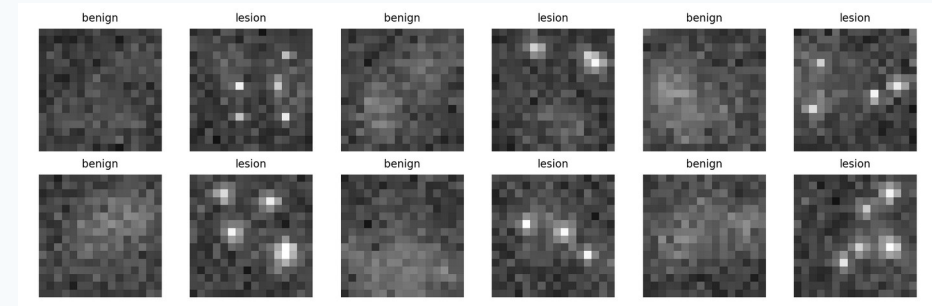
Safe claim: 100% on a 40-sample held-out prototype set - not clinical accuracy and not IBM hardware.



Problem & Motivation

Early disease signals are subtle, noisy, and label-limited.

- Microscopic pathology patterns can appear as tiny texture changes, weak margins, or small lesion clusters.
- Deep CNNs can perform well, but they usually need more labeled data and many trainable parameters.
- Hackathon target: prove an end-to-end, reproducible quantum-classical diagnostic workflow.

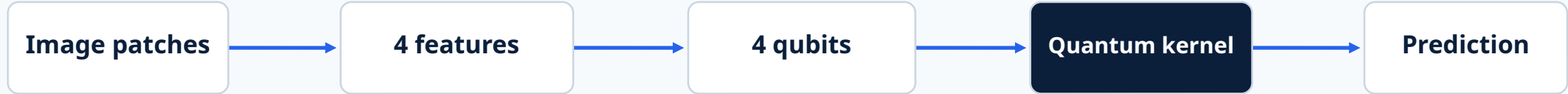


Clinical framing

Classify benign vs early-lesion patches, then retrieve similar cases for explanation.

Proposed Solution: End-to-End Workflow

A compact pipeline that judges can run and inspect in minutes.



Validation layer: held-out metrics + 5-fold CV + ideal/noisy quantum-kernel benchmark

Retrieval layer: Pinecone or local JSON stores compressed diagnostic embeddings and case metadata.

Runnable demo: `python -m qbio_diagnostics.run_demo --samples 160 --qubits 4`

Quantum Implementation

Technical heart: ZZ-style feature map + fidelity kernel + SVM decision boundary.

```
feature_map = zz_feature_map(feature_dimension=4, reps=1,
                             entanglement="full")
quantum_kernel = FidelityQuantumKernel(feature_map=feature_map)
model = QSVC(quantum_kernel=quantum_kernel)
```

- Kernel: $K(x_i, x_j) = |\langle \phi(x_i) | \phi(x_j) \rangle|^2$
- Classifier: SVM trained over the quantum similarity matrix
- IBM hardware path is optional and separately labeled as not_run until a QPU job completes.

Representation

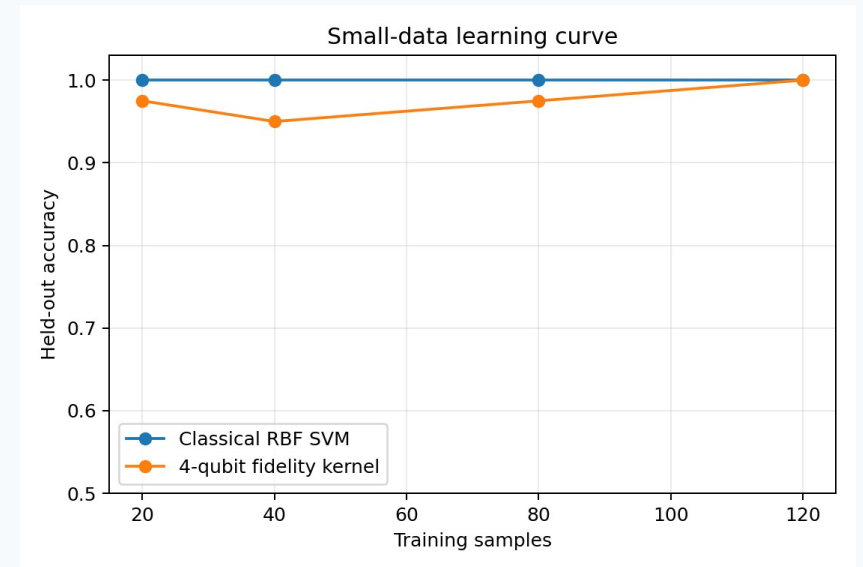
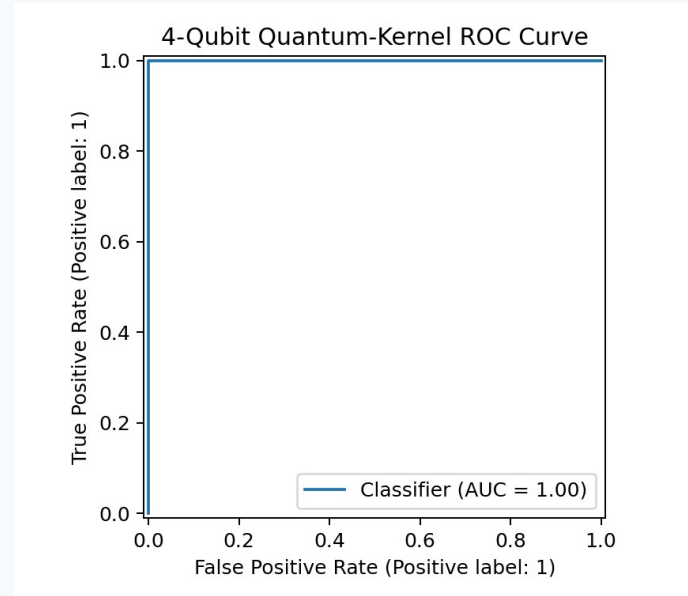
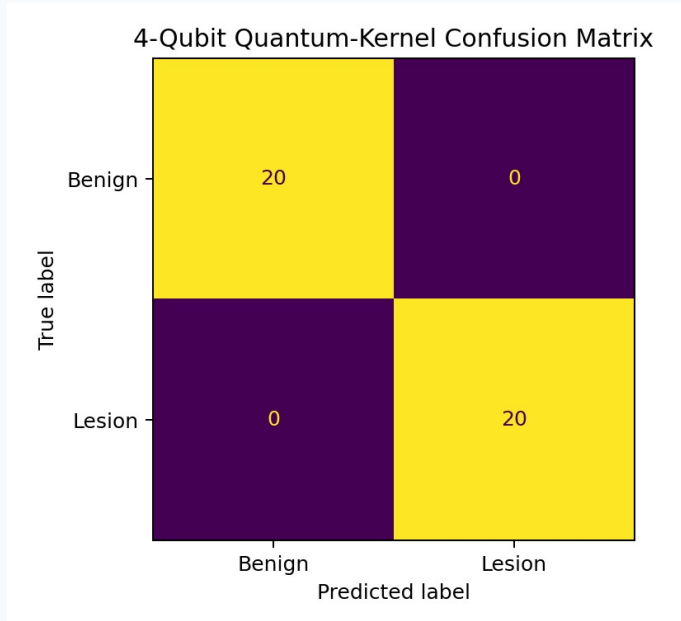
Each 16x16 patch is compressed into 4 normalized descriptors: std, max intensity, bright ratio, local contrast.

Fallback honesty

If Qiskit is unavailable, the project uses a 4-qubit reference quantum-kernel simulator. It does not silently switch to a classical polynomial kernel.

Results & Verification

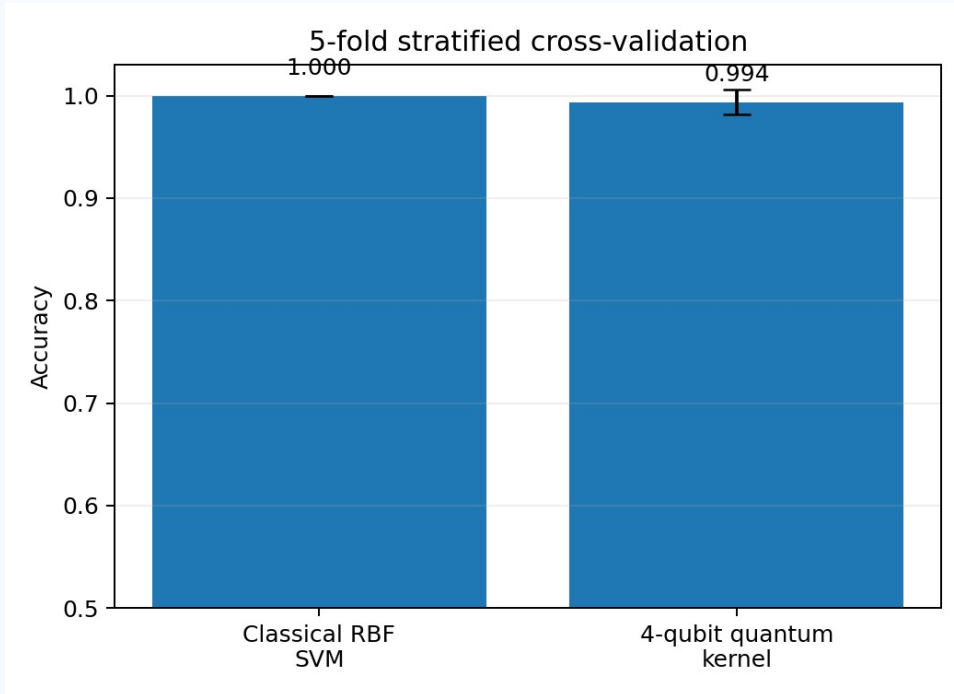
Prototype metrics from 160 reproducible biomedical-style patches.



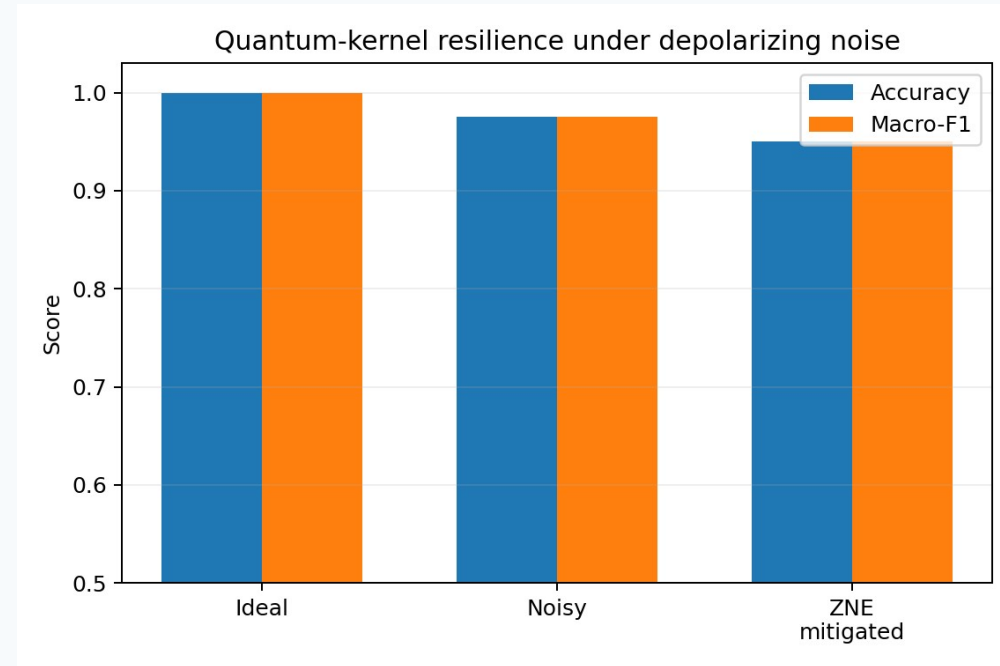
Say this: quantum-kernel evaluation achieved 100% accuracy on the 40-sample held-out evaluation set. Do not call it clinical accuracy or IBM hardware accuracy.

Robustness: Cross-Validation + Noise

The updated version adds the evidence judges are likely to ask for.



5-fold quantum-kernel CV mean accuracy: 0.9938



Noisy classifier accuracy in stress test: 0.975

IBM Quantum hardware status: not_run in packaged result; the repo includes an optional hardware benchmark starter.

Future Scope & Submission Readiness

Ready for submission: repo + deck + demo-video script.

- Upload the full folder to a public GitHub repo with README, requirements, notebook, and outputs.
- Record a 2-4 minute video showing terminal execution, metrics JSON, and output charts.
- In the pitch, separate prototype simulation results from optional future IBM QPU execution.

Roadmap

1. Replace synthetic data with BreakHis/HAM10000/X-ray patches
2. Run IBM Runtime Sampler/Estimator jobs
3. Add readout mitigation and hardware-specific calibration
4. Build clinician review workflow

Final pitch line

Q-BioDiag turns tiny biomedical image signals into quantum-kernel similarity patterns, tests robustness under noise, and retrieves explainable diagnostic cases.